Negative Swap Spreads: A Simple Explanation

Understanding why "risky" rates fell below "safe" Treasury yields

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What is a Swap Spread?

Think of it as a simple math problem that compares two interest rates:

Swap Spread = Swap Rate - Treasury Yield

Same maturity (e.g., both 30-year)

EXAMPLE CALCULATION

30-Year Swap Rate

3.0%

Rate banks charge each other (based on LIBOR)

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30-Year Treasury Yield

2.5%

Risk-free rate from U.S. government bonds

=

Swap Spread

+50 bps

Positive spread (normal)

Why SHOULD Swap Spreads Be Positive?

Imagine you're lending money to two borrowers:

OPTION 1

U.S. Government

(via Treasury bonds)

Rock solid - will definitely pay

OPTION 2

Big Banks

(via interest rate swaps, based on LIBOR)

Riskier - banks can fail

Who would you charge a higher interest rate?

The banks! Because they're riskier than the U.S. government.

Swap rates should be HIGHER
than Treasury yields

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Swap spreads should be POSITIVE
Historically: +30 to +60 bps

SEPTEMBER 2008

The Anomaly Begins: Lehman Collapses

Within weeks of the financial crisis, something bizarre happened to 30-year swap spreads:

BEFORE LEHMAN

+40 bps

Normal positive spread

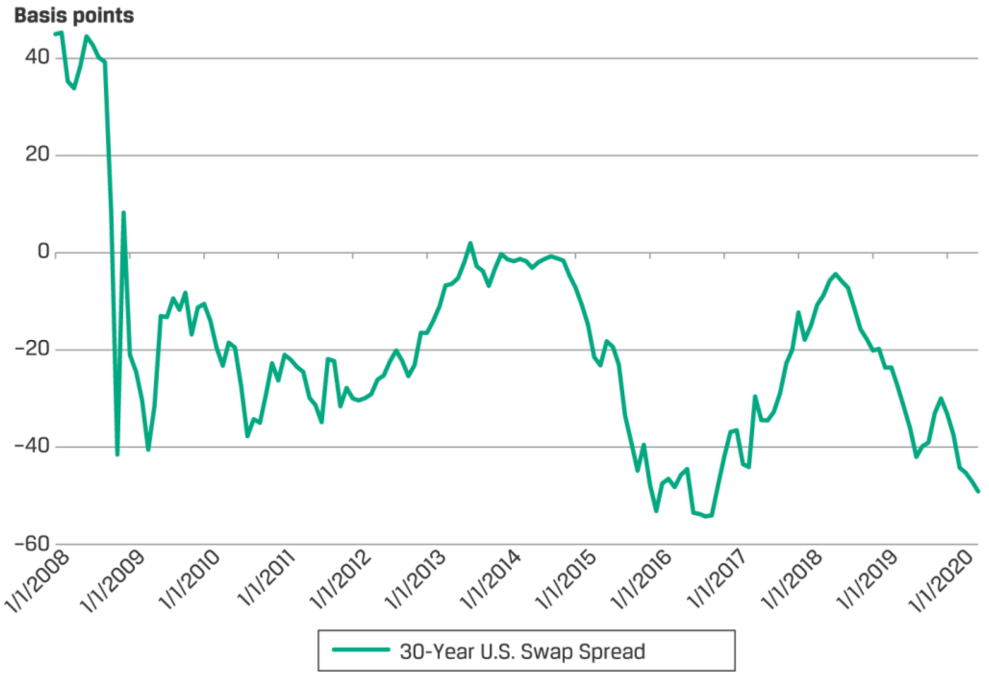
NOVEMBER 2008

-62 bps

Inverted! Makes no sense

The "risky" rate fell BELOW the "safe" rate

This violates basic financial logic!



30-Year U.S. Swap Spread (2008–2020)

Source: CFA Level II Curriculum

REASON 1

Pension Funds Desperately Needed Duration

Pension funds promised to pay retirees billions in 30 years. When rates fell, those promises became MORE expensive. They needed "duration" - assets that go up when rates fall.

A Buy 30-Year Treasuries

Need to pay $1 billion CASH upfront. Most pension funds were already underfunded - they didn't have billions lying around!

Too expensive

vs

B Enter 30-Year Swap

Only need 2-5% margin (like $30-50M deposit). Gets the SAME duration exposure as buying the Treasury!

Much cheaper!

⚠ What Happened After Lehman Collapsed

1. Stock market crashed - assets fell
2. Rates fell - liabilities increased
3. Pension funds massively underfunded
4. All rushed to receive fixed in swaps

THE RESULT

Everyone wanted to receive fixed, pushing swap rates DOWN below Treasury yields

REASON 2

Dealers Couldn't Arbitrage It Away

Normally, if spreads went negative, smart traders (dealers) would do this "free money" trade:

The "Arbitrage" Trade

1. **Pay fixed on the swap**
   at 2.38%
2. **Buy the Treasury bond**
   yielding 3.00%

✓ **Net profit = 0.62% per year**
Free money, right?

Not in 2008!

The normal "arbitrage police" who would fix the pricing broke down completely during the crisis.

What Went Wrong

Funding Markets Froze
Dealers couldn't borrow money cheaply to buy Treasuries. Liquidity dried up overnight.

Counterparty Risk
Who wants to enter a 30-year swap when banks are failing? Trust evaporated.

Risk-Off Panic
Everyone wanted to sell risky assets, including dealers who might have done this trade.

2008-2016: WHY IT STAYED NEGATIVE

The New Sheriff: Supplementary Leverage Ratio

After 2008, regulators said: "Banks took too much risk! We need MORE capital requirements." They introduced the SLR.

What is the SLR?

THE RULE

For every $100 of assets, hold $5-6 in capital

Your own equity as a buffer against losses

Sounds reasonable, right? But here's the problem:

The Problem

Treasury bonds count as FULL assets even though they're super safe. No credit for being risk-free!

How This Killed the Arbitrage Trade

BEFORE SLR

Capital Required: ~1%

Dealers could do the trade with minimal capital. Easy arbitrage, spreads would normalize quickly.

WITH 6% SLR

Capital Required: ~6%

6x more capital needed! The trade became uneconomical. Dealers left the "free money" on the table.

The Math: Why Dealers Stopped Arbitraging

Same trade, $10 million notional. See how the return changed:

THE OLD DAYS

Before SLR

Trade Size	$10,000,000
Capital Required	$100,000 (1%)
Annual Profit	$37,490

RETURN ON EQUITY

37%

THE NEW WORLD

With 6% SLR

Trade Size	$10,000,000
Capital Required	$643,800 (6%)
Annual Profit	$37,490

RETURN ON EQUITY

6%

Dealers want 15% ROE minimum. At 6% SLR, spreads need to be -54 bps to make 15% ROE!

The "free money" isn't worth picking up anymore.

Putting It All Together

2008

How We Got Here

1. Lehman Brothers collapsed, triggering financial crisis
2. Pension funds panicked, needed duration hedging fast
3. Flooded market wanting to receive fixed in 30-year swaps
4. Dealers couldn't arbitrage due to crisis conditions

Swap spreads went negative

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Post 2008

Why It Stayed That Way

1. Pension funds stayed underfunded (stocks slow, rates low)
2. They kept needing to receive fixed in swaps
3. New SLR regulations made arbitrage expensive
4. Dealers needed -50 to -70 bps to make it worth it

No arbitrage = spreads stayed broken

The Bottom Line

THINK OF IT LIKE THIS

"Imagine there's a $20 bill lying on the sidewalk, but you need to rent a $500 crane to pick it up because of new city regulations. You're not going to bother, right?"

That's exactly what happened with negative swap spreads:

There was "free money" (negative spreads)

But regulations made it too expensive to pick up

So dealers left it there until 2016

The pricing anomaly persisted because the cost of fixing it was too high given regulatory constraints.

References

NY FED PAPER

"Negative Swap Spreads"

Nina Boyarchenko, Pooja Gupta, Nick Steele, Jacqueline Yen

Federal Reserve Bank of New York Economic Policy Review, Vol. 24, No. 2, October 2018

Explains how regulatory leverage ratios (SLR) reduced dealer incentives to arbitrage negative spreads, creating a "new normal" in swap pricing.

BIS WORKING PAPER

"An Explanation of Negative Swap Spreads: Demand for Duration from Underfunded Pension Plans"

Sven Klingler, Suresh Sundaresan

BIS Working Papers No. 705, Bank for International Settlements, February 2018

Shows how underfunded pension funds' demand for duration hedging via swaps drove the initial inversion of swap spreads in 2008.

CFA CURRICULUM CFA Institute Level II Curriculum — Fixed Income: Swap Spreads (Chart: 30-Year U.S. Swap Spread 2008-2020)